

$$\begin{aligned}
 \text{Since, } f(\theta) &= \frac{1}{1 + \tan \theta} \\
 \therefore f(\alpha) \cdot f(\beta) &= \frac{1}{1 + \tan \alpha} \times \frac{1}{1 + \tan \beta} \\
 &= \frac{1}{1 + \tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta} \\
 &= \frac{1}{1 + 1} \\
 &= \frac{1}{2}
 \end{aligned}$$

48. Option (b) is correct.

$$\begin{aligned}
 \tan \alpha + \tan \beta &= 6 \\
 \tan \alpha \cdot \tan \beta &= 8 \\
 \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} \\
 &= \frac{6}{1 - 8} = \frac{6}{-7} \\
 \cos(2\alpha + 2\beta) &= \cos 2(\alpha + \beta) \\
 &= \frac{1 - \tan^2(\alpha + \beta)}{1 + \tan^2(\alpha + \beta)} \\
 &= \frac{1 - \frac{36}{49}}{1 + \frac{36}{49}} \\
 &= \frac{13}{85}
 \end{aligned}$$

49. Option (c) is correct.

$$\begin{aligned}
 \tan(90 - 25)^\circ + 2 - 2 \tan 40^\circ - \tan 25^\circ \\
 &= \cot 25^\circ - \tan 25^\circ - 2 \tan 40^\circ + 2 \\
 &= \frac{\cos^2 25 - \sin^2 25}{\sin 25 \cdot \cos 25} - 2 \tan 40^\circ + 2 \\
 &= \frac{2 \cos 50^\circ}{\sin 50^\circ} - 2 \tan 40^\circ + 2 \\
 &= 2 \cot 50^\circ - 2 \tan 40^\circ + 2 \\
 &= 2 \tan 40^\circ - 2 \tan 40^\circ + 2 = 2
 \end{aligned}$$

50. Option (a) is correct.

If triangle is acute angled triangle then

$$\cot A \cdot \cot B \cdot \cot C > 0$$

So, statement 1 is correct.

If triangle is obtuse angled triangle, then one of A, B, C is obtuse i.e., one of the $\tan A$, $\tan B$ and $\tan C$ is -ve

$$\therefore \tan A \cdot \tan B \cdot \tan C < 0$$

So, statement 2 is wrong.

51. Option (a) is correct.

Given the circle of the equation is

$$x^2 + y^2 + 2x + 6y + 1 = 0$$

To find the centre (a, b) and the radius c of a circle equation, we need to rewrite the equation in the form $(x-h)^2 + (y-k)^2 = r^2$, where (h, k) is the centre of the circle and r is its radius.

$$\text{Let } x^2 + y^2 + 2x + 6y + 1 = 0$$

First complete the square to get the equation in the desired form:

$$(x^2 + 2x + 1) - 1 + (y^2 + 6y + 9) - 9 + 1 = 0$$

$$\Rightarrow (x+1)^2 - 1 + (y+3)^2 - 9 + 1 = 0$$

$$\Rightarrow (x+1)^2 + (y+3)^2 - 9 = 0$$

$$\Rightarrow (x+1)^2 + (y+3)^2 = 9$$

$$\Rightarrow [x - (-1)]^2 + [y - (-3)]^2 = (3)^2$$

Now we can see that the centre of the circle is at (-1, -3) and the radius is 3.

$$\therefore a = -1, b = -3 \text{ and } c = 3$$

$$\begin{aligned}
 \therefore a^2 + b^2 + c^2 &= (-1)^2 + (-3)^2 + (3)^2 \\
 &= 1 + 9 + 9 \\
 &= 19
 \end{aligned}$$

52. Option (a) is correct.

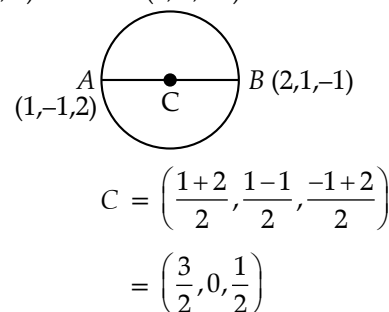
The general equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a sphere with centre

$$(-u, -v, -w) \text{ and radius } r = \sqrt{u^2 + v^2 + w^2 - d}$$

If B is the mid-point of the line segment AC where $A = (x_1, y_1, z_1)$ and $C = (x_2, y_2, z_2)$ then

$$B = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Given: Equation of sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz - 1 = 0$ with AB as its diameter and $A = (1, -1, 2)$ and $B = (2, 1, -1)$



$$\begin{aligned}
 C &= \left(\frac{1+2}{2}, \frac{-1+1}{2}, \frac{2-1}{2} \right) \\
 &= \left(\frac{3}{2}, 0, \frac{1}{2} \right)
 \end{aligned}$$

As we know that centre of the sphere is given by: $(-u, -v, -w)$

So compare that we get

$$\left(\frac{3}{2}, 0, \frac{1}{2} \right) = (-u, -v, -w)$$

$$u = -\frac{3}{2}, v = 0, w = -\frac{1}{2}$$